

Human-imperceptible Privacy Protection Against Machines

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ABSTRACT

Privacy concerns with social media have recently been under the spotlight, due to a few incidents on user data leakage on social networking platforms. With the current advances in machine learning and big data, computer algorithms often act as a first-step filter for privacy breaches, by automatically selecting content with sensitive information, such as photos that contain faces or vehicle license plate. In this paper we propose a novel algorithm to protect the sensitive attributes against machines, meanwhile keeping the changes imperceptible to humans. In particular, we first conducted a series of human studies to investigate multiple factors that influence human sensitivity to the visual changes. We discover that human sensitivity is influenced by multiple factors, from low-level features such as illumination, texture, to high-level attributes like object sentiment and semantics. Based on our human data, we propose for the first time the concept of *human sensitivity map*. With the sensitivity map, we design a human-sensitivity-aware image perturbation model, which is able to modify the computational classification results of sensitive attributes while preserving the remaining attributes. Experiments on real world data demonstrate the superior performance of the proposed model on human-imperceptible privacy protection.

CCS CONCEPTS

• **Information systems** → **Multimedia information systems**;
• **Human-centered computing** → **Visualization**; • **Computing methodologies** → *Computer vision*.

KEYWORDS

Privacy protection, social multimedia, deep neural network

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	Original image		Perturbed image	
				
	Human judgment	Machine judgment	Human judgment	Machine judgment
People present?	✓	✓	✓	✗
Text present?	✓	✓	✓	✗
Contain focused object?	✓	✓	✓	✓
Aesthetic?	✓	✓	✓	✓
Pleasant?	✓	✓	✓	✓

Figure 1: We propose a human-sensitivity-aware image perturbation model for privacy protection against machines. Our model is able to prevent machines from learning sensitive scene-level attributes (e.g., “people present”, “text present”) while preserving others (e.g., “image aesthetics”), meanwhile keeping the perturbation imperceptible to humans.

1 INTRODUCTION

In recent years, we have witnessed an explosive growth in the social media community. Massive amounts of data are shared online, leading to substantial privacy issues and security concerns. The situation is worsened with the recent breakthroughs in machine learning, in which online data can be processed automatically for inferring sensitive information, such as personal identities, social relationships, and the type of environment an individual is interacting with. Indeed, machines are now acting as an efficient filter for many privacy attacks, especially in the field of online media. We advocate the transfer of emphasis on privacy protection against humans to privacy protection against machines. That is, we need to prevent machine learning algorithms to learn from online data automatically. Meanwhile, although our target is to fool machines, it is also important to keep the changes in images imperceptible

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Figure 2: Sample images in our stimulus set. From left to right: original image, tampered image with noise added on the human body, tampered image with noise on the plate, tampered image with noise on background.

to humans. This is to maintain the image quality, aesthetics, and function, especially in the scenario of online data sharing in social media platforms.

Many studies report adversarial perturbation methods that suppress machine predictions for images [23, 36, 41, 42, 62]. A common approach is to embed human-imperceptible noise in images. However, as the algorithms change the images as a whole, the modification will alter other attributes as well. Furthermore, although most of the above studies aim to make their changes imperceptible to humans, they do not elaborate what image factors influence human visual sensitivity.

Researchers have been striving to address the above issues. A recent study [14] proposes an algorithm for anonymizing selected attributes for face images while keeping the remaining face attributes unchanged. The work in [13] proposes to use GAN-generated faces instead of real faces. However, existing work focuses only on face images, whereas we notice that privacy leakage does not only occur for faces, or not even only for objects, but also on image context. New algorithms on privacy protection on scene-level attributes are highly needed.

For human imperceptibility, some researchers incorporated saliency map in their perturbation models [25, 47, 65], by hypothesizing that observers are most sensitive to the changes in regions that garner most human attention. While such measures have brought considerable improvement in performance, little is known about the relation between visual saliency and visual sensitivity.

In this study, we empirically explore the impact of multiple level of image features on human visual sensitivity, and use the human findings to design a sensitivity-aware image perturbation model that prevents machines from learning sensitive attributes (see Fig. 1). For human studies, we add Gaussian noise with a sigma of 0.06 to each object as well as the background in an image (see Fig. 2), and ask observers to distinguish the tampered image from the original image (see Fig. 3). Additive Gaussian noise is often used in the research of just noticeable difference (JND) for evaluating human perception to image distortion [29, 31, 39, 49]. We identify multiple levels of object features that influence human sensitivity, from low-level features such as object texture, to mid-level features like object size and shape, and high-level features including object sentiment, object semantics, and object saliency. We find that image context also influences human sensitivity. Building on these insights we propose a new concept—*sensitivity map*, which shows different levels of human sensitivity within an image. With the sensitivity map, we design a sensitivity-aware image perturbation model that

is able to modify the sensitive image attributes (e.g., having people in the scene), while keeping other attributes unaltered (e.g., the visual aesthetics). On a real world dataset, the proposed model demonstrates its advantage on privacy protection for images of general scenes. Follow-up human studies show that the perturbed images from the proposed model are more imperceptible to humans than those from baseline (i.e., the model without sensitivity map), suggesting the efficacy of human sensitivity integration.

Our contributions are summarized as follows.

- We perform a comprehensive analysis on the factors impacting human visual sensitivity. We find that human sensitivity is influenced by multiple object features, such as size, texture, and sentiment, as well as the image context. Our analyses expand an existing dataset with ground truth human sensitivity score for each image.
- We propose a new concept—*sensitivity map*, which represents different levels of human sensitivity within an image, along with a deep neural network (DNN) for automated sensitivity map generation. For the first time we measure quantitatively the relation between human visual sensitivity and visual saliency.
- We develop a sensitivity-aware image perturbation model for privacy protection against machines. The proposed model is able to prevent machines from learning sensitive attributes while keeping other attributes unaltered. By integrating the image sensitivity map, the proposed algorithm is able to reach a higher human imperceptibility level than the baseline model without sensitivity map.

2 RELATED WORK

2.1 Privacy Protection for Visual Data

In recent years privacy protection for visual data has been becoming a topic of immense interest [6, 8, 18, 20, 53, 54]. Traditional approaches mainly focus on detecting and removing sensitive objects (e.g., faces, license plates) [19, 21, 52, 55]. Two recent studies have extended face and license plates to a wider set of objects (e.g., nudity, hand writing) and have applied deep neural network for detection and recognition tasks [44, 45]. Most of the existing models strive to maintain the visual quality while protecting the privacy [34, 35, 48]. Our work is built on the previous literature but we extend the object-level sensitive attributes to scene-level attributes, such as “people present in the scene” and “text present in the scene”.

For human imperceptibility, we perform a comprehensive analysis to explore how human sensitivity is influenced by image features.

2.2 Adversarial Learning for Privacy Protection

With the recent development of generative adversarial networks (GANs), several works have proposed GAN-generated adversarial examples that can fool a classifier to predict wrong classes [27, 32, 47, 50, 61]. Coming to the visual domain, the existing research is mainly limited to face images, which usually includes a noise-generating network that is able to alter the sensitive face attributes, such as gender, race, and age [40, 51, 60]. One latest example is from Chhabra and colleagues [14], who propose an adversarial perturbation based algorithm to anonymize selected face features while keeping other features unaltered. Cheung et. al use GAN-generated faces for training face classifiers, in the aim of protecting the real face data [13].

To make the changes human imperceptible, a few researchers have incorporated visual saliency in adversarial perturbation or detection [33, 64, 65]. Our work is clearly distinct from the above. First, we work on images of general scenes instead of solely faces. Second, we adopt a broader set of sensitive attributes, such as having people or text in the scene. We propose a new concept, namely *sensitivity map* and show that visual sensitivity is not the same as visual saliency. Our computational experiments show that integrating the sensitivity map into the DNN model significantly improves human imperceptibility of the image perturbation.

2.3 Just Noticeable Difference in Images

In the area of experimental psychology, a just-noticeable difference (JND) is the amount something must be changed in order for a difference to be noticeable, detectable at least half the time [22]. Researchers have adopted this notion to account for the maximum sensory distortion that the typical human visual system does not perceive (e.g., for 75% of the observers), for applications in perceptual image/video processing algorithms and systems. For example, the work in [58] introduces a pixel-wise JND estimation model for images based upon evaluation of pixel orientations and pattern complexity. Jakhetiya and colleagues [31] proposes to measure pixel-based JND based on contrast sensitivity by measuring root mean square contrast.

Although there has been a plethora of work on predicting JND for images [31, 39, 49, 59], most of them focus on the low-level image features such as contrast [31], edges [39, 56], and texture [39]. Much less work has been done on the relation between JND and mid- and high-level image attributes, such as object semantics, and image aesthetics. Our research is motivated by the above studies but we go further. In particular, we design human experiments to test the relation between human sensitivity and object- and scene-level features, such as object size, shape, sentiment, image aesthetics, and so on. Compared to traditional bottom-up JND, our sensitivity map incorporates top-down human cognition characteristics and takes into account both object-level and scene-level context.

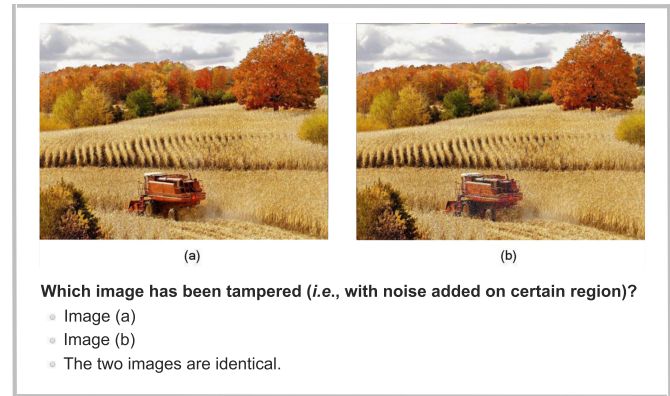


Figure 3: The graphical interface of the online human study on visual sensitivity.

3 EMPIRICAL STUDIES

We perform empirical human studies to explore how multiple image factors influence observer sensitivity. This is done through psychophysics experiments on Amazon Mechanical Turk (MTurk) [46]. Below we describe the human studies design and psychophysics data analyses.

3.1 Experiment Paradigm

Stimuli: We used the EMotional attention dataset (EMOd) [24] as our stimuli. EMOd contains 1019 emotional images each with 1) detailed object-level labels for sentiment (negative, neutral, positive) and semantic categories (neutral face, emotional face, objects with motion, objects with sound, etc. See the supplementary materials for details); 2) 33 scene-level attributes such as “aesthetics” and “attractive person” (see the supplementary material); 3) eye fixation data collected from 16 human subjects. Due to copyright issues, we exclude the photos from the International Affective Picture System (IAPS) [37] for online human studies, resulting in 860 image stimuli in total.

In order to evaluate the impact of multiple image factors on human perception sensitivity, we added Gaussian noise with a sigma of 0.06 on each object in each image. We added noise on the background (i.e., the region without any object labels) also (see Fig. 2 for examples). In order to fit in the computer screen for human experiment, each image was resized to 384×512 pixels.

Design and procedure: To measure human sensitivity, we presented workers on MTurk with a game titled “Can you spot the difference?”. Each participant saw 60 pairs of images sequentially. Each pair contains one original image from EMOd and a test image. The test image could be either tampered (with noise added on a particular object or background) or intact (i.e., same as the original image). The images in each pair were presented to participants side by side. Each pair was only shown for 10 seconds. A notice was given before the display of each pair. Participants could click the “play” button to view the images when they were ready. Participants were to decide which image in the pair was tampered by answering a multiple choice question below the image (see Fig. 3).

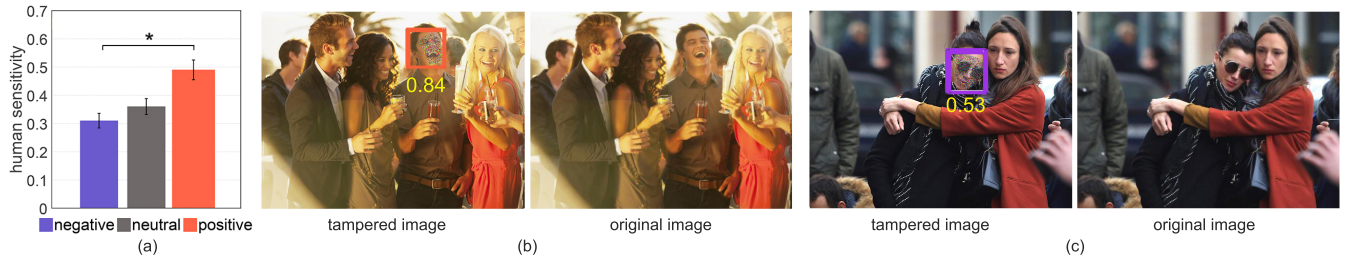


Figure 4: (a) Human sensitivity is higher on positive objects than on negative objects. The error bars indicate the standard error of the mean. The asterisks are denoted as following: * $p < .05$, ** $p < .01$, * $p < .001$. (b–c) Images illustrates how human sensitivity is higher on positive face (outlined in red) than on negative face (outlined in purple). In each pair, the tampered image is on the left and the original image is on the right. The number below each rectangle denotes human sensitivity score of the outlined region.**

The whole experiment paradigm was based on previous literature on just noticeable difference in images [31, 58].

Analytical approach: In total we had 234 MTurk workers (157 male, age: 30.18 ± 5.32) participating in the human experiments. On average each image was judged by 9 workers. The average accuracy of the 9 responses (ranges between 0 and 1 with higher value indicates higher sensitivity) was stored as the human sensitivity score to each image. We analyzed the relation between human sensitivity and object- and scene-level features. For categorical attributes like object sentiment, we first performed univariate analyses of variance (ANOVA's) by setting human sensitivity as the dependent variable. If an ANOVA was significant, we followed up with post hoc Tukey test to see where differences were. For continuous attributes (e.g., object size), we used Spearman's rank correlation to test its relation between human sensitivity. These tests are standard in behavioral and other sciences. See, for example, [4] for an introduction to inferential statistics.

3.2 The Impact of Object-level Attributes on Human Sensitivity

In this subsection, we quantify the degree to which human sensitivity can be explained by various object-level attributes. The analyses are performed in three levels: 1) low-level object features including object color, illumination, texture [30]; 2) mid-level objects features including object size, convexity, solidity, and complexity [63]; 3) high-level object attributes including object sentiment and object semantics [24]; 4) object saliency (i.e., object attention score—the maximum fixation-map value that is inside the object's contour [24]). Previous literature have demonstrated the high impact on the size of the noise region on human perception [5, 7, 28], and our analysis also indicates a strong correlation between object size and human sensitivity (Spearman's $\rho = 0.68$, $p < 0.0001$), thus we normalize the human sensitivity by object size in the following analyses.

A series of Spearman's rank correlation show that one low-level features (object histogram/texture) and two mid-level object features (object convexity and object solidity) moderately correlate with human sensitivity (Spearman's $|\rho| \geq 0.16$, $p < 0.01$ ¹, see Fig.

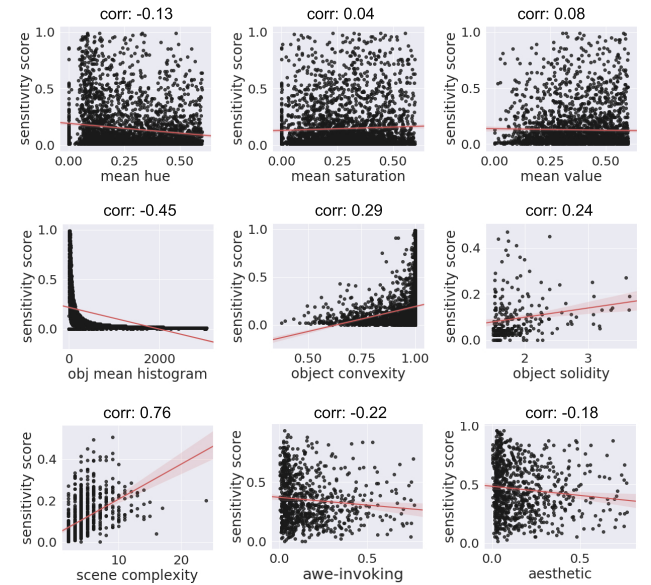


Figure 5: The correlation between human sensitivity and object- and scene-level image features. Low-level object features do not significantly affect human sensitivity except pixel histogram. Human sensitivity is strongly influenced by object pixel histogram (i.e., texture) and scene complexity (indicated by number of objects in an image). The number on each chart denotes the Spearman's correlation ρ .

5), indicating that object appearance (i.e., size, texture, shape) play a part in tricking human eyes. No significant correlations were found on object color, illumination and object complexity (Spearman's $\rho \leq 0.05$, $p > 0.05$).

Interestingly, we find that the two types of high-level object attributes significantly influence human sensitivity. An ANOVA indicates that object sentiment significantly influences human sensitivity ($F(2, 3270) = 3.74$, $p = .024$ ²). A post hoc Tukey test shows

²We report the results of ANOVAs as, " $F(df_{condition}, df_{error}) = F$ value, $p = p$ value". If a p value is smaller than the conventional significance level threshold of 0.05, we reject the null hypothesis of no difference among the means.

¹Throughout the paper, ρ s, p s represents the plural form of ρ , p , respectively.

that observers have higher sensitivity on positive objects than negative objects ($p = 0.018$, see Fig. 4). Regarding semantics, human sensitivity is highest on emotional faces, followed by neutral faces and watchable objects (see the supplementary materials for details).

Not surprisingly, analyses show a strong correlation between human sensitivity and object attention score (Spearman's $\rho = 0.63$, $p < 0.0001$), suggesting that visual saliency impacts human sensitivity.

We further compare human sensitivity on objects with that on image background. We computed the average accuracy on all objects within an image to present the object accuracy for the image. Paired sample t -test show that human is more sensitive on objects than on background (average sensitivity score: 0.68 v.s. 0.62, $t(859) = 5.31$, $p < 0.001^3$), suggesting that human sensitivity is lower on background than on objects.

3.3 The Impact of Image Context on Human Sensitivity

Image context has been known to influence various human visual processes, such as attention [26], memorability [9], and object recognition [17, 43]. In this section, we show that human sensitivity is not only impacted by the object features to which the noise is added, it is also affected by the image context as a whole.

First of all, we represent image complexity in three ways: 1) scene complexity (*i.e.*, number of objects within an image; 2) sentiment complexity (*i.e.*, types of object sentiments within an image; and 3) semantics complexity (*i.e.*, types of object semantic categories within an image). We represent human sensitivity for a particular image using the averaged accuracy of all objects in it. As shown in Fig. 5, human sensitivity moderately and negatively correlates with scene and sentiment complexity (Spearman's $|\rho|s \geq 0.43$, $ps < 0.0001$), and weakly correlate with semantics complexity (Spearman's $\rho = -0.16$, $p < 0.01$).

We select images that contain 6 to 11 objects to minimize the effect of image complexity, and analyze the relation between human sensitivity and other scene-level attributes, which include 1) low-level image features including image entropy, texture, color and illumination; 2) mid-level image features which is represented by the entropy of human attention map; 3) 33 high-level perceptual image attributes provided in EMOd. Results show that the overall image illumination weakly and positively correlates with human sensitivity (Spearman's $\rho = 0.18$, $ps < 0.01$), suggesting that higher illumination is helpful for increasing human sensitivity. The entropy of human attention map weakly and negatively correlates with human sensitivity (Spearman's $\rho = -0.19$, $ps < 0.01$), which indicates that less focused images lead to lower sensitivity.

Human sensitivity is also influenced by high-level image attributes. As shown in Fig. 5, the more aesthetic, awe-invoking or unusual an image is, the lower human visual sensitivity. On the contrast, when it comes to image layout, the closer-up shot and having more focused object improves human sensitivity. For images that contain people, the more attractive and interacting the persons are in the image, the higher observer sensitivity will be.

³We report the results of t -tests as, “ $t(df) = t$ value, $p = p$ value”.

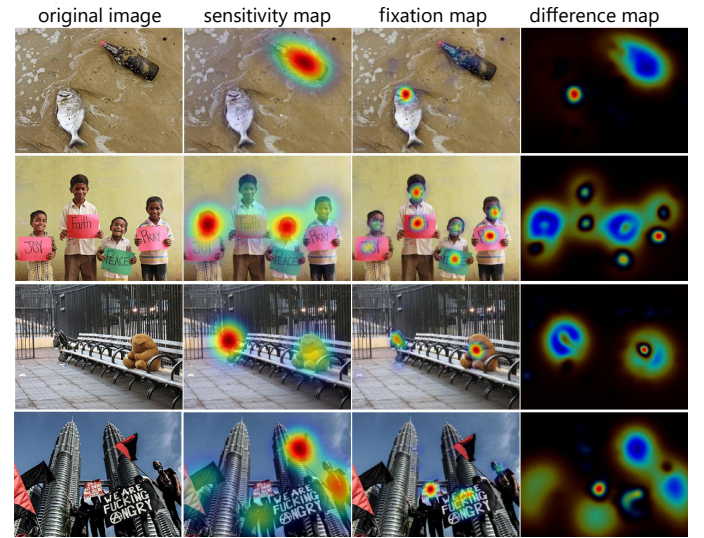


Figure 6: The regions that garner most human attention might not necessarily overlap with the regions of highest human sensitivity. We visualize the difference between sensitivity map and fixation map. The last column (difference image) visualizes the difference between the two types of maps.

3.4 The Relation between Human Sensitivity and Visual Saliency

Previous studies have used human saliency map to guide the adversarial attack [25, 47, 65]. In this subsection, we show both quantitative and qualitative difference between human sensitivity and visual saliency.

A one-way ANOVA shows that whether an object is being focused or not significantly impacts human sensitivity on it ($F(1, 3270) = 55.62$, $p < .00001$). Spearman's correlation between each object's sensitivity and attention score indicates moderate and positive correlation between these two variables ($\rho = 0.43$, $ps < 0.01$). These results suggest that an object's saliency enhances human visual sensitivity towards it.

Despite the above correlation, we observed fundamental differences between human sensitivity and visual saliency. First of all, the impact of object size is much stronger on human sensitivity ($\rho = 0.68$, $ps < 0.01$) than on visual saliency ($\rho = 0.39$, $ps < 0.01$). Second, object sentiment influences more on human attention ($F(2, 3270) = 62.71$, $p < .00001$) than on human sensitivity ($F(2, 3270) = 3.74$, $p = .0024$). Notably, both positive and negative objects attract human attention more than neutral objects, whereas only positive object enhances human sensitivity compared with neutral and negative objects. In fact, negative objects degrades human sensitivity. Object semantics have marginally stronger effect size on human sensitivity ($F(13, 3270) = 13.26$, $p < .00001$) than on visual saliency ($F(13, 3270) = 12.37$, $p < .00001$). The biggest difference is on watchable objects (*e.g.*, screen)—humans have high sensitivity to noise on this type objects but have lower attention score.

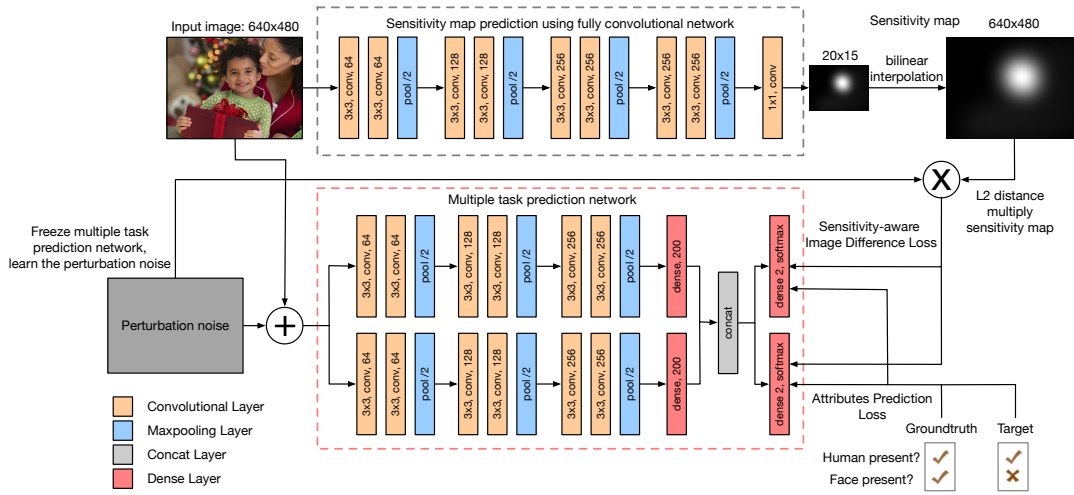


Figure 7: Overview of the proposed sensitivity-aware image perturbation model. The network consists of two branches: Sensitivity map generation branch (outlined in dotted gray line in the first row) and sensitivity-aware image perturbation branch (outlined in dotted red line below).

We further generate a sensitivity map for each image, similar to the way for generating psychophysical fixation maps [38, 57, 66]. More specifically, for each subject i viewing object j , assuming that each subject’s judgment gives rise to a Gaussian-distributed activity, the sensitivity map (S_I) for image I are represented as the recorded judgment convolved with an isotropic Gaussian kernel K_G , and then normalizing the map to maximum 1, as below

$$S_I = \alpha \sum_{i=1}^n s_i K_G M_j \quad (1)$$

where s_i represents the sensitivity score of the i th subject out of the n subjects. M_j denotes the object mask which defines the image region, and α normalizes the map.

To quantitatively evaluate the difference between sensitivity map and fixation map, we adopt two image similarity metrics, namely Pearson’s Correlation Coefficient (CC) and histogram intersection (SIM), which are commonly used in the saliency community [12]. We get a CC of 0.47 and SIM of 0.43, suggesting that there are substantial differences between these two types of maps [10, 12]. Fig. 6 shows some qualitative examples.

In summary, our human studies suggest that human sensitivity is a complex phenomenon that is influenced by both bottom-up and top-down cognitive processes. Our findings on the impacts of color and illumination are consistent with previous studies of JND that texture and illumination affects human sensitivity [16, 39]. Our findings of the impact of image context echo previous studies which report that image context influences human attention [3, 26]. We extend the previous work to higher-level image features including sentiment, semantics, and visual saliency. Our results provide a more nuanced understanding of human sensitivity to image distortion, and guide the computational models for human-imperceptible privacy protection, which are discussed in the following sections.

4 PRIVACY PROTECTION AGAINST MACHINES

In this section, we propose a sensitivity-aware image perturbation model for the protection of sensitive image attributes against machines. We demonstrate that with the help of sensitivity map, the proposed algorithm is able to make human imperceptible changes on the images, so as to prevent machines from learning targeted attributes while preserving other attributes.

4.1 Experimental Settings

Datasets: The proposed algorithm is evaluated on the author collected subset of EMOd [24], which includes 860 emotional images depicting general real life scenes.

DNN architecture: The architecture of our DNN is illustrated in Fig. 7. This proposed model consists of two branches: (1) sensitivity map generation branch and (2) sensitivity-aware image perturbation branch. For sensitivity map generation branch, we adopt traditional fully convolutional network based on VGG-16, followed by a 1×1 convolutional layer to combine all feature maps extracted from VGG-16, to get the final sensitivity map. We first feed the original image with resolution $640 \times 480 \times 3$ into our network to get a map with a resolution of 20×15 , we then apply bilinear interpolation to get the full-size sensitivity map (640×480).

For sensitivity-aware image perturbation branch, we first train a multiple task model as proposed in [2] to predict multiple scene-level attributes, such as “human present”, “aesthetic”. We then set a learnable noise and add it to the original image as the perturbed image. Next, we feed this perturbed image into the trained multiple task model for noise learning. There are two different losses in this model: Attributes prediction loss and image difference loss.

For attributes prediction loss, we first separate all attributes into two different parts: attributes to be suppressed (e.g., sensitive attributes that have potential privacy leakage risk, such as “human

present” and “face present”) and attributes to be preserved ((e.g., “aesthetic”, “pleasant”). In the multiple task attribute prediction model, we are able to get the logits of original image and perturbed image. The loss is designed based on these logits. We apply L1 loss for all attributes. We minimize and maximize the L1 loss for attributes to be preserved and those to be suppressed, respectively.

For image difference loss, we integrate image sensitivity information into the L2 loss, by introducing the sensitivity map values as per-pixel weights when computing weighted L2 norms of the per-pixel difference between the original and the perturbed image. Our design is different from prior work [14], which directly adopts L2 distance on original image and perturbed image, whereas we take into account the relative importance of different regions within the image. As a result, the image regions that human is sensitive to receive less perturbation.

In order to integrate the knowledge of human sensitivity, we extend the objective function in [14] by multiplying the sensitivity map with the regularization term, as shown below:

$$\min \{D(I_{Ap}, P_{Ap}) - D(I_{As}, P_{As}) + \beta_s \|I - T\|_2^2\} \quad (2)$$

where, D is the distance metric. A_s and A_p denote the sets of attributes to be suppressed and preserved where, $A \geq (A_s \cap A_p)$ and $A_s \cup A_p = \emptyset$. I and P represent the original image and the perturbed image, respectively. β_s is the factor extracted from the sensitivity map.

The training and testing procedures are on a single NVIDIA 1080 Ti GPU using Keras with a Tensorflow backend [1, 15]. For sensitivity map prediction network, we initialize the training to the pretrained parameters for VGG-16 on ImageNet. The parameters of the DNN are then learned in an end-to-end fashion on the EMOd dataset with stochastic gradient descent. The learning rate is 0.01, momentum is 0.9 and weight decay is 0.0005 for 30 epochs. For attributes prediction network, we train the network from scratch on EMOd dataset with the attributes to be suppressed. The optimizer is stochastic gradient descent. The learning rate is 0.01, momentum is 0.9 and weight decay is 0.0005. We run 20 epochs for this training. Then we iteratively modify the prediction of the attributes to be suppressed while preserving the remaining attributes. Our algorithm is extended from CW-attack, we use Adam optimizer to learn the noise. The learning rate is 0.01, beta1 is 0.9 and beta2 is 0.999. The max-iteration is 10000. If the max-iteration is reached, the image attack is considered being failed. The EMOd dataset is divided into three different folds to obtain more statistically meaningful results by applying cross-validation. The whole process takes approximately 2 hours.

4.2 Performance Evaluation

The proposed model is tested for single and multiple attributes anonymization. More specifically, the algorithm needs to ensure that the class of A_s attributes in image I changes to some other classes while preserving A_p attributes in the same image. We ran two experiments on EMOd. First, we set two sensitive image attributes for A_s , namely “human present” and “text present”. The A_p set includes “contain focused object”, “aesthetic”, and “pleasant”. In our second experiment, we set “human present” as A_s and “face present” as A_p . We then compare our model with baseline (i.e., the model without sensitivity map, which is original proposed in [14]).

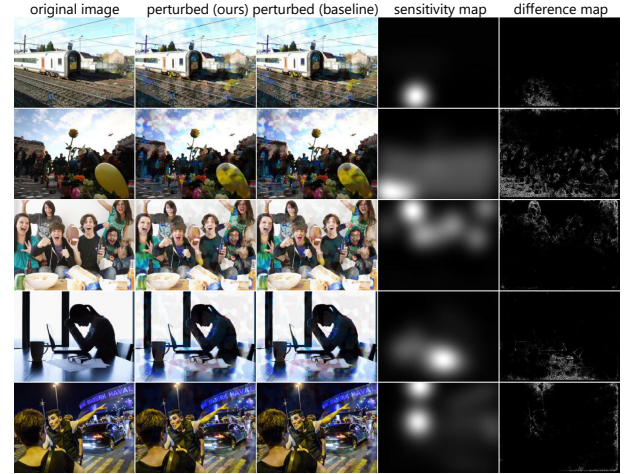


Figure 8: Comparisons of the perturbed image generated by the proposed model and that from baseline. The last column is the difference between the two perturbed images on each row. The perturbed images by the proposed model have significant less noise on the most sensitive region, suggesting the efficacy of the integration of sensitivity map.

The performances are evaluated on three aspects: 1) The efficacy of sensitivity map; 2) targeted attributes suppression and other attributes preservation; 3) human imperceptibility.

The efficacy of sensitivity map: First of all, we evaluate the performance of sensitivity map prediction. We get a CC of 0.47 and SIM of 0.59 between the predicted sensitivity map and human ground truth, indicating an acceptable performance on sensitivity prediction [11]. For each perturbed image generated by the proposed model, we calculate its difference between the perturbed image from the baseline. We use L2 distance to transfer the three channel difference image to grayscale, and compare with the ground truth human sensitivity map. We show in Fig. 8 example images containing people or text, and compare the difference between the perturbed images generated by the proposed model with those from baseline. Results show that significant less noise is added on the most sensitive region for our perturbed images, suggesting the efficacy of the integration of sensitivity map.

Attributes suppression and preservation: The results are shown in Table 1 and Fig. 9. We see that the predictions of the targeted attributes change significantly after perturbation, whereas the prediction of the other attributes remain largely unaltered. This suggests that our model is able to prevent machines from learning specific sensitive attributes while preserving the remaining ones. The confusion matrix of baseline shows similar performance (see the supplementary material), suggesting that integrating sensitivity map did not impair the privacy protection performance. We further modify the perturbation level by changing the learning rate and maximum iteration rate, and see how attributes prediction and image quality vary. Results show that perturbation level does not significantly influence performance (see the supplementary material for details).

Table 1: Confusion matrix displaying classification accuracy (%) of five attributes. Two attributes (“people present” and “text present”) are suppressed, while three attributes (“focused object”, “aesthetic” and “pleasant”) are preserved. The 2×2 numbers below each attributes denote (in clockwise order starting from upper left) true positive, false positive, false negative, and true negative, respectively.

Attributes	Suppressed attributes				Preserved attributes					
	People Present		Text Present		Focused Object		Aesthetic		Pleasant	
Before perturbation	0.996	0.004	0.916	0.084	0.998	0.002	0.897	0.103	0.946	0.054
	0.082	0.918	0.000	1.000	0.195	0.805	0.000	1.000	0.000	1.000
After perturbation	0.252	0.748	0.210	0.790	0.998	0.002	0.921	0.079	0.979	0.021
	0.857	0.143	0.752	0.248	0.180	0.820	0.002	0.998	0.009	0.991

Human imperceptibility: We perform a online human study to evaluate the difference on human imperceptibility to the perturbed images generated by the proposed model and baseline model. The experiment paradigm is the same as introduced in Section 3.1. We select all images from EMOd that contain either face, human, or text, resulting in a subset of 346 images. In total 80 MTurk workers participated the human study and each image is judged by 9 participants on average.

Similar to Section 3.1, we compute the average human sensitivity score for each image. Paired sample t -test shows that human sensitivity is significantly lower on perturbed images generated by the proposed model than those by the baseline (average sensitivity score: 0.60 v.s. 0.62, $t(688) = 3.33$, $p < 0.001$), suggesting that the perturbation by our model is more imperceptible to humans.

In summary, from three different aspects we demonstrate the efficacy of the proposed model for privacy protection for images of general scenes. By integrating human sensitivity map, our model is able to achieve better human imperceptibility while maintaining the same perturbation function.

4.3 Limitations and future work

One major limitation of our proposed model is detection-method specific, that is, it mainly works for the model which has been used for the perturbation training. We will take the problem for more generalised perturbation as our future work. Ideally, the method should be robust against non-DNN based approaches as well. However, we foresee that as we make our model more generalized to both deep learning and non-DNN based methods, preserving visual imperceptibility will become more challenging.

Furthermore, we designated the sensitive attributes based on previous literature on privacy [44, 45]. In fact, the whole scene could also be sensitive, such as the scene inside a hospital. One of the following work is to perturb the automated scene recognition.

5 CONCLUSIONS

In this research, we work on human-imperceptible privacy protection against machines. We perform extensive human studies to empirically analyze the impact of both low- and high-level image features on human visual sensitivity. For the first time, we propose the concept of *human sensitivity map* that visualizes different levels of human sensitivity to the visual changes within an image. Based on the human studies we propose a sensitivity-aware image perturbation model that is able to fool machines on targeted attributes while preserving others. Experiments on images of general scenes

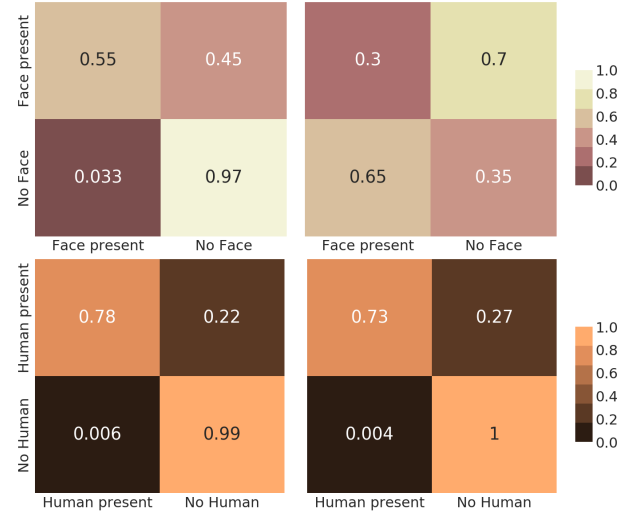


Figure 9: Confusion matrix displaying classification accuracy (%) on the two sensitive attributes before and after perturbation. Our target is to suppress the attribute “face present” and preserve the attribute “human present”. The 2×2 numbers in each block denotes (in clockwise order starting from upper left) true positive, false positive, false negative, and true negative, respectively.

demonstrate the model’s efficacy on private attributes protection while keeping the changes imperceptible to humans. The proposed sensitivity map can be applied in numerous applications, such as image retouching and computer graphics rendering.

In the future, we plan to extend our analysis to privacy protection in videos. We also plan to re-structure the attributes prediction network by taking into consideration more factors such as attributes correlation and conflict.

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REFERENCES

- [1] Martin Abadi, Ashish Agarwal, Paul Barham, Eugene Brevdo, Zhifeng Chen, Craig Citro, Greg S Corrado, Andy Davis, Jeffrey Dean, Matthieu Devin, et al. 2016. Tensorflow: Large-scale machine learning on heterogeneous distributed systems. *arXiv preprint arXiv:1603.04467* (2016).

- [2] Abrar H Abdunabi, Gang Wang, Jiwen Lu, and Kui Jia. 2015. Multi-task CNN model for attribute prediction. *IEEE Transactions on Multimedia* 17, 11 (2015), 1949–1959.
- [3] Tamar Avraham and Michael Lindenbaum. 2010. Esaliency (extended saliency): Meaningful attention using stochastic image modeling. *IEEE transactions on pattern analysis and machine intelligence* 32, 4 (2010), 693–708.
- [4] R.A. Bailey. 2008. *Design of comparative experiments*. Vol. 25. Cambridge University Press.
- [5] Peter GJ Barten. 1999. *Contrast sensitivity of the human eye and its effects on image quality*. Vol. 21. Spie optical engineering press Bellingham, WA.
- [6] Michael Boyle, Carman Neustaedter, and Saul Greenberg. 2009. Privacy factors in video-based media spaces. In *Media Space 20+ Years of Mediated Life*. Springer, 97–122.
- [7] Antoni Buades, Bartomeu Coll, and J-M Morel. 2005. A non-local algorithm for image denoising. In *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05)*, Vol. 2. IEEE, 60–65.
- [8] Monika Büscher, Sung-Yueh Perng, and Michael Liegl. 2019. Privacy, Security, and Liberty: ICT in Crises. In *Censorship, Surveillance, and Privacy: Concepts, Methodologies, Tools, and Applications*. IGI Global, 199–217.
- [9] Zoya Bylinskii, Phillip Isola, Constance Bainbridge, Antonio Torralba, and Aude Oliva. 2015. Intrinsic and extrinsic effects on image memorability. *Vision research* (2015).
- [10] Zoya Bylinskii, Tilke Judd, Ali Borji, Laurent Itti, Frédo Durand, Aude Oliva, and Antonio Torralba. [n. d.]. MIT Saliency Benchmark.
- [11] Zoya Bylinskii, Tilke Judd, Frédo Durand, Aude Oliva, and Antonio Torralba. 2015. Mit saliency benchmark.
- [12] Zoya Bylinskii, Tilke Judd, Aude Oliva, Antonio Torralba, and Frédo Durand. 2019. What do different evaluation metrics tell us about saliency models? *IEEE transactions on pattern analysis and machine intelligence* 41, 3 (2019), 740–757.
- [13] Sen-Ching Samson Cheung, Herb Wildfeuer, Mehdi Nikkhah, Xiaoqing Zhu, and Waitian Tan. 2018. Learning Sensitive Images Using Generative Models. In *2018 25th IEEE International Conference on Image Processing (ICIP)*. IEEE, 4128–4132.
- [14] Saheb Chhabra, Richa Singh, Mayank Vatsa, and Gaurav Gupta. 2018. Anonymizing k-facial attributes via adversarial perturbations. *IJCAI* (2018).
- [15] François Chollet. 2015. Keras. *GitHub repository* (2015).
- [16] Chun-Hsien Chou and Yun-Chin Li. 1995. A perceptually tuned subband image coder based on the measure of just-noticeable-distortion profile. *IEEE Transactions on circuits and systems for video technology* 5, 6 (1995), 467–476.
- [17] Xiao Chu, Wei Yang, Wanli Ouyang, Cheng Ma, Alan L Yuille, and Xiaogang Wang. 2017. Multi-context attention for human pose estimation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 1831–1840.
- [18] Serdar Çiftçi, Ahmet Öğüz Akýüz, and Touradj Ebrahimi. 2018. A reliable and reversible image privacy protection based on false colors. *IEEE transactions on Multimedia* 20, 1 (2018), 68–81.
- [19] Rita Cucchiara. 2005. Multimedia surveillance systems. In *Proceedings of the third ACM international workshop on Video surveillance & sensor networks*. ACM, 3–10.
- [20] Ling Du, Wei Zhang, Huazhu Fu, Wenqi Ren, and Xinpeng Zhang. 2019. An Efficient Privacy Protection Scheme for Data Security in Video Surveillance. *Journal of Visual Communication and Image Representation* (2019).
- [21] Frederic Dufaux and Touradj Ebrahimi. 2008. Scrambling for privacy protection in video surveillance systems. *IEEE Transactions on Circuits and Systems for Video Technology* 18, 8 (2008), 1168–1174.
- [22] GÖsta Ekman. 1959. Weber's law and related functions. *The Journal of Psychology* 47, 2 (1959), 343–352.
- [23] Gamaleldin Elsayed, Shreya Shankar, Brian Cheung, Nicolas Papernot, Alexey Kurakin, Ian Goodfellow, and Jascha Sohl-Dickstein. 2018. Adversarial Examples that Fool both Computer Vision and Time-Limited Humans. In *Advances in Neural Information Processing Systems*. 3914–3924.
- [24] S. Fan, Z. Shen, M. Jiang, B. Koenig, J. Xu, M. Kankanhalli, and Q. Zhao. 2018. Emotional Attention: A Study of Image Sentiment and Visual Attention. In *Computer Vision and Pattern Recognition (CVPR), 2018 IEEE Conf. on*.
- [25] Ruth C Fong and Andrea Vedaldi. 2017. Interpretable explanations of black boxes by meaningful perturbation. In *Proceedings of the IEEE International Conference on Computer Vision*. 3429–3437.
- [26] Stas Goferman, Lihi Zelnik-Manor, and Ayelet Tal. 2012. Context-aware saliency detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 34, 10 (2012), 1915–1926.
- [27] Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. 2015. Explaining and harnessing adversarial examples. *ICLR* (2015).
- [28] Hongmei Gou, Ashwin Swaminathan, and Min Wu. 2007. Noise features for image tampering detection and steganalysis. In *2007 IEEE International Conference on Image Processing*. Vol. 6. IEEE, VI–97.
- [29] Walid Hachicha, Azeddine Beghdadi, and Faouzi Alaya Cheikh. 2013. Stereo image quality assessment using a binocular just noticeable difference model. In *2013 IEEE International Conference on Image Processing*. IEEE, 113–117.
- [30] Laurent Itti, Christof Koch, and Ernst Niebur. 1998. A model of saliency-based visual attention for rapid scene analysis. *IEEE Transactions on Pattern Analysis & Machine Intelligence* 11 (1998), 1254–1259.
- [31] Vinit Jakhethiya, Weisi Lin, Sunil Jaiswal, Ke Gu, and Sharath Chandra Guntuku. 2018. Just Noticeable Difference for natural images using RMS contrast and feed-back mechanism. *Neurocomputing* 275 (2018), 366–376.
- [32] Felix Juefei-Xu, Vishnu Naresh Boddeti, and Marios Savvides. 2018. Perturbative neural networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 3310–3318.
- [33] Dimitris Kastaniotis, Ioanna Ntinou, Dimitrios Tsourounis, George Economou, and Spiros Fotopoulos. 2018. Attention-aware generative adversarial networks (ATA-GANs). In *2018 IEEE 13th Image, Video, and Multidimensional Signal Processing Workshop (IVMSP)*. IEEE, 1–5.
- [34] Pavel Korshunov, Shuting Cai, and Touradj Ebrahimi. 2012. Crowdsourcing approach for evaluation of privacy filters in video surveillance. In *Proceedings of the ACM multimedia 2012 workshop on Crowdsourcing for multimedia*. ACM, 35–40.
- [35] Pavel Korshunov and Touradj Ebrahimi. 2013. Using face morphing to protect privacy. In *2013 10th IEEE International Conference on Advanced Video and Signal Based Surveillance*. IEEE, 208–213.
- [36] Alexey Kurakin, Ian Goodfellow, and Samy Bengio. 2017. Adversarial machine learning at scale. *ICLR* (2017).
- [37] Peter J Lang, Margaret M Bradley, and Bruce N Cuthbert. 2008. International affective picture system (IAPS): Affective ratings of pictures and instruction manual. *Technical report A-8* (2008).
- [38] Olivier Le Meur and Thierry Baccino. 2013. Methods for comparing scanpaths and saliency maps: strengths and weaknesses. *Behavior research methods* 45, 1 (2013), 251–266.
- [39] Anmin Liu, Weisi Lin, Manoranjan Paul, Chenwei Deng, and Fan Zhang. 2010. Just noticeable difference for images with decomposition model for separating edge and textured regions. *IEEE Transactions on Circuits and Systems for Video Technology* 20, 11 (2010), 1648–1652.
- [40] Vahid Mirjalili, Sebastian Raschka, Anoop Namboodiri, and Arun Ross. 2018. Semi-adversarial networks: Convolutional autoencoders for imparting privacy to face images. In *2018 International Conference on Biometrics (ICB)*. IEEE, 82–89.
- [41] Seyed-Mohsen Moosavi-Dezfooli, Alhussein Fawzi, Omar Fawzi, and Pascal Frossard. 2017. Universal adversarial perturbations. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 1765–1773.
- [42] Seyed-Mohsen Moosavi-Dezfooli, Alhussein Fawzi, and Pascal Frossard. 2016. Deepfool: a simple and accurate method to fool deep neural networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2574–2582.
- [43] Aude Oliva, Antonio Torralba, Monica S Castelano, and John M Henderson. 2003. Top-down control of visual attention in object detection. In *Image processing, 2003. icip 2003. proceedings. 2003 international conference on*, Vol. 1. IEEE, 1–253.
- [44] Tribhuvanesh Orekondy, Mario Fritz, and Bernt Schiele. 2018. Connecting pixels to privacy and utility: Automatic redaction of private information in images. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 8466–8475.
- [45] Tribhuvanesh Orekondy, Bernt Schiele, and Mario Fritz. 2017. Towards a visual privacy advisor: Understanding and predicting privacy risks in images. In *Proceedings of the IEEE International Conference on Computer Vision*. 3686–3695.
- [46] Gabriele Paolacci, Jesse Chandler, and Panagiotis Ipeirotis. 2010. Running experiments on amazon mechanical turk. *Judgment and Decision Making* 5, 5 (2010), 411–419.
- [47] Nicolas Papernot, Patrick McDaniel, Xi Wu, Somesh Jha, and Ananthram Swami. 2016. Distillation as a defense to adversarial perturbations against deep neural networks. In *2016 IEEE Symposium on Security and Privacy (SP)*. IEEE, 582–597.
- [48] So-hyeon Park and Kyung-ho Lee. 2017. Developing Criteria for Invasion of Privacy by Personal Drone. In *2017 International Conference on Platform Technology and Service (PlatCon)*. IEEE, 1–7.
- [49] Chuan Qin, Chin-Chen Chang, and Chia-Chun Lin. 2015. An adaptive reversible steganographic scheme based on the just noticeable distortion. *Multimedia Tools and Applications* 74, 6 (2015), 1983–1995.
- [50] Konda Reddy Mopuri, Utkarsh Ojha, Utsav Garg, and R Venkatesh Babu. 2018. NAG: Network for adversary generation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 742–751.
- [51] Aria Rezaei, Chaowei Xiao, Jie Gao, and Bo Li. 2018. Protecting Sensitive Attributes via Generative Adversarial Networks. *arXiv preprint arXiv:1812.10193* (2018).
- [52] Slobodan Ribaric, Aladdin Ariyaeinia, and Nikola Pavesic. 2016. Identification for privacy protection in multimedia content: A survey. *Signal Processing: Image Communication* 47 (2016), 131–151.
- [53] Shabnoor Siddiqui, Tajinder Singh, et al. 2016. Social media its impact with positive and negative aspects. *International Journal of Computer Applications Technology and Research* 5, 2 (2016), 071–075.
- [54] Junjie Wang, Brandon Amos, Anupam Das, Padmanabhan Pillai, Norman Sadeh, and Mahadev Satyanarayanan. 2017. A scalable and privacy-aware IoT service for live video analytics. In *Proceedings of the 8th ACM on Multimedia Systems Conference*. ACM, 38–49.

- [55] Junjue Wang, Brandon Amos, Anupam Das, Padmanabhan Pillai, Norman Sadeh, and Mahadev Satyanarayanan. 2018. Enabling Live Video Analytics with a Scalable and Privacy-Aware Framework. *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)* 14, 3s (2018), 64.
- [56] Shiqi Wang, Lin Ma, Yuming Fang, Weisi Lin, Siwei Ma, and Wen Gao. 2016. Just noticeable difference estimation for screen content images. *IEEE Transactions on Image Processing* 25, 8 (2016), 3838–3851.
- [57] David S Wooding. 2002. Fixation maps: quantifying eye-movement traces. In *Proceedings of the 2002 symposium on Eye tracking research & applications*. ACM, 31–36.
- [58] Jinjian Wu, Leida Li, Weisheng Dong, Guangming Shi, Weisi Lin, and C-C Jay Kuo. 2017. Enhanced just noticeable difference model for images with pattern complexity. *IEEE Transactions on Image Processing* 26, 6 (2017), 2682–2693.
- [59] Jinjian Wu, Guangming Shi, Weisi Lin, Anmin Liu, and Fei Qi. 2013. Just noticeable difference estimation for images with free-energy principle. *IEEE Transactions on Multimedia* 15, 7 (2013), 1705–1710.
- [60] Zhenyu Wu, Zhangyang Wang, Zhaowen Wang, and Hailin Jin. 2018. Towards Privacy-Preserving Visual Recognition via Adversarial Training: A Pilot Study. In *Proceedings of the European Conference on Computer Vision (ECCV)*. 606–624.
- [61] Chaowei Xiao, Bo Li, Jun-Yan Zhu, Warren He, Mingyan Liu, and Dawn Song. 2018. Generating adversarial examples with adversarial networks. *arXiv preprint arXiv:1801.02610* (2018).
- [62] Cihang Xie, Jianyu Wang, Zhishuai Zhang, Yuyin Zhou, Lingxi Xie, and Alan Yuille. 2017. Adversarial examples for semantic segmentation and object detection. In *Proceedings of the IEEE International Conference on Computer Vision*. 1369–1378.
- [63] Juan Xu, Ming Jiang, Shuo Wang, Mohan S Kankanhalli, and Qi Zhao. 2014. Predicting human gaze beyond pixels. *Journal of vision* 14, 1 (2014), 28–28.
- [64] Fuxun Yu, Qide Dong, and Xiang Chen. 2018. ASP: A Fast Adversarial Attack Example Generation Framework based on Adversarial Saliency Prediction. *arXiv preprint arXiv:1802.05763* (2018).
- [65] Chiliang Zhang, Zuochang Ye, Yan Wang, and Zhimou Yang. 2018. Detecting Adversarial Perturbations with Saliency. In *2018 IEEE 3rd International Conference on Signal and Image Processing (ICSIP)*. IEEE, 271–275.
- [66] Qi Zhao and Christof Koch. 2011. Learning a saliency map using fixated locations in natural scenes. *Journal of vision* 11, 3 (2011), 9–9.